DISPLAYS

LCD GRADES

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Have you ever heard of LCD grades? Chances are you have not, and this is a feature that's actually getting four turds for being missing...not for existing! Let us explain.

If you've ever bought a high-end monitor, or an expensive TV you might actually have come across the term LCD Grade. This is because it's a feature used (sometimes) to market screens to those in the know. However, it's something that most manufacturers are quite content not letting the general public in on. Look at the table below, and you will understand that when LCDs are manufactured, they're tested and graded. A, B and C are the most common grades at manufacturing, though you might come across A+ grading very often for high end products.

better contrast ratio produces a more realistic image. In practice though, there is no industry standard way of measuring the contrast ratio, or even a standard definition of what a contrast ratio is. In any case, what a contrast ratio should be is what the actual LCD panel is capable of displaying, but manufacturers do not put these actual numbers on the box. Instead, they use terms such as "dynamic contrast ratio", or "advanced contrast ratio", or even just "contrast ratio". Then there is a number that looks something like 100,000,000:1. The number is entirely arbitrary, and is indicating something other than the native contrast ratio of the panel. As Digit readers you're already aware of how to find real native contrast ratios of panels, so we won't take up space explaining it again. But the fact that many still fall for these meaningless numbers earns it the highest turd rating.

LCD GRADES	A	B	C
Dead Pixels	3	5	8
Marks	None	Unnoticeable	Allowed
Blemishes	None	Unnoticeable	Allowed
Scratches	None	Allowed on edges	Allowed
Backlight leaking	None	None	Allowed

The LCD grading system

The way it works is simple. If you buy yourself a cheap monitor or TV, you're likely getting a B or C grade panel. Most often it isn't a hugely noticeable difference, but it is a difference nonetheless. Remember, if you spread this knowledge, and make sure everyone you know never buys a panel that doesn't give the grade information, you can make enough of a difference to maybe start forcing them to do so.

DYNAMIC CONTRAST RATIO

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Contrast ratios are another specification where manufacturers are trying to outdo each other to sell their products. A contrast ratio is essentially the blackest part of an image in relation to the lightest part of the image. In theory, a

INBUILT GAMES

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We know right! Team Digit actually giving anything to do with gaming a turd? It's almost like you entered the *Twilight Zone*! Anyway, the games that are built into TVs are usually played with the remote. Usually they're something as simple as a scrolling racing game, or a *Tetris* or *Pacman* ripoff. Rarely, you might find match-3 or bubble bursting games. To add to it, some TV manufacturers try and sell you joysticks to be able to play the game better. Most people don't shell out extra and end up using their remotes. Now, paying extra for this feature is pointless, because you're likely to never use it, and more often than not, your family is going to tell you to put some channel on instead of playing a silly game that you could do on your phone anyway. Overall, we think it's a total waste of time and effort, and we can't believe that it's some poor people's job to build these pointless things! Besides, if you do

This might actually convince us to choose the model.

play them, remember you will probably screw up your remote, and have to shell out a bomb for a new one!

REFRESH RATES

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Flatscreen manufacturers are no different from every other technology category, and they also believe that bumping up the specs makes their products stand out. One of the most abused specs is the refresh rate. We now get TVs boasting of 120Hz or 240Hz, and insisting in their advertising that this translates to smoother viewing. Problem is, normal TV and DVDs, etc are running at 24 to 30 frames per second (fps). One fps = 2 Hz refresh rate, thus 30 fps = 60 Hz. Remember that your signal is coming in at 60 Hz, so if you want to run something at 120 Hz, it's essentially repeating images in quick succession. It can be worse. The TV might have some code built in that delays your signal ever so slightly from displaying, so that it can read ahead and try to give you an approximation of a middle frame, instead of just repeating a frame twice. This can often make the picture look worse! Of course, the refresh rate technology isn't bad by itself. It's the fact that there's just no content to utilise 120 to 240 Hz. Games might be able to, but then there's the continued debate on whether humans can really enjoy the difference between 120 and 240 Hz.